

Differentiable functions – 1

Derivatives: computation, growth and behaviour at infinity

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Theoretical minimum.

Def. f is differentiable at x_0 if $f(x) = f(x_0) + f'(x_0)(x - x_0) + o(x - x_0)$; differentiability is strictly stronger than continuity, and the derivative itself need not be continuous.

Theorem 1 (rules for higher derivatives). Leibniz' rule $(fg)^{(n)} = \sum_k \binom{n}{k} f^{(k)} g^{(n-k)}$; for a rational function the n -th derivative is found from the partial fraction decomposition, since $\left(\frac{1}{x-a}\right)^{(n)} = \frac{(-1)^n n!}{(x-a)^{n+1}}$. Coefficients of a power series give derivatives at the centre: $f^{(n)}(0) = n! c_n$.

Theorem 2 (behaviour at infinity). If f is differentiable on (a, ∞) and $f(x) + f'(x) \rightarrow L$, then $f(x) \rightarrow L$ and $f'(x) \rightarrow 0$; the same trick — multiply by e^x and apply l'Hôpital to $\frac{e^x f(x)}{e^x}$ — settles most problems of this type.

Fact 1. A bounded function with $ff' \geq g$ cannot converge if $\int g$ oscillates without bound: differentiating f^2 turns a statement about f into a statement about the primitive of g . This is the engine of Problem 3.

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1. (VJIMC 1992) Find the n -th derivative of $f(x) = \frac{x}{x^2 - 1}$.
 2. (VJIMC 1997) Let $c_1, \dots, c_n$ be real numbers with $c_1^k + c_2^k + \dots + c_n^k > 0$ for every $k = 1, 2, \dots$, and put $f(x) = \frac{1}{(1 - c_1 x)(1 - c_2 x) \cdots (1 - c_n x)}$. Show that $f^{(k)}(0) > 0$ for all $k = 1, 2, \dots$.
 3. (VJIMC 2013) Let $f: [0, \infty) \rightarrow \mathbb{R}$ be differentiable with $|f(x)| \leq M$ and $f(x)f'(x) \geq \cos x$ for $x \geq 0$. Prove that $f(x)$ has no limit as $x \rightarrow \infty$.
 4. (VJIMC 2014) Let $f: (0, \infty) \rightarrow \mathbb{R}$ be differentiable and suppose $\lim_{x \rightarrow \infty} (f(x) + \frac{f'(x)}{x}) = 0$. Prove that $\lim_{x \rightarrow \infty} f(x) = 0$.

5. (IMC 2014) Let $f(x) = \frac{\sin x}{x}$ for $x > 0$ and let n be a positive integer. Prove that $|f^{(n)}(x)| < \frac{1}{n+1}$.